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MILITARY SPECIFICATION
PROJECTILE, TP-T, 152MM, XM411E1
METAL PARTS ASSEMBLY

1. SCOPE

* 1.1 This specification covers Projectile, TP-T, 152MM, XM411E1, Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-C-60539	- Coatings, Anodic, Conventional, for Aluminum and Aluminum Alloys
MIL-R-11073	- Rotating Band Blanks, Sintered Iron

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metal, Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

FSC 1320

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2.1 (Cont'd)

MIL-STD-1167	- Ammuntiion Data Card
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

F10534387	- Projectile, TP-T, 152MM, XM411E1, Metal Parts Assembly
F8889328	- Pallet for Fixed and Semifixed Projectiles
IEL-10534387	- Inspection Equipment List for Projectile, TP-T, 152MM, XM411E1, Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Heat treatment or stress relief.-The heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in any component the mechanical properties specified on the applicable drawings.

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3.2.1 Hot forged-heat treat method of manufacture (if applicable).-Projectile components manufactured by this method shall be subjected to heat treatment to attain required mechanical properties. The heat treatment of ferrous forgings shall consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit (F), followed by an oil quench. After quenching, the piece shall be tempered without delay at a temperature above 700 degrees F. suitable to produce required mechanical properties. After removal from tempering furnace, the projectile components shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 degrees F. after which cooling may be accelerated by water spray or similar means. Nonferrous forgings shall be heat treated to standard commercial practice for the particular alloy to attain required mechanical properties.

3.2.2 Cold shaping method of manufacture (if applicable).-Cold shaping of steel components shall be so accomplished as to produce in the projectile component the mechanical properties specified. The projectile component shall be stress relieved at a temperature of 700 degrees F., minimum, following the final forming operation during which time plastic flow takes place. The minimum soak time at or above minimum temperature shall be established which will produce mechanical properties specified by the applicable drawing. Nonferrous cold shaped components shall be heat treated after forming to attain required mechanical properties.

3.3 Rotating band (sintered iron).

3.3.1 Band seat.-Prior to application of rotating band blank, the band seat of the body shall be clean and free of oil, grease, dirt, rust and other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seat after knurling.

3.4 Protective coating.-Protective coatings shall be in accordance with the applicable drawings and the specifications referenced thereon. Where protective coating is required by component details, the requirements will apply to all indicated surfaces regardless of contour configuration.

3.5 Painting.-Painting shall comply with the requirements of the applicable drawings. All paint shall be dry to the touch before testing and packing for shipment.

* 3.6 Proving ground functional tests.-The assembled projectile metal parts shall reveal no evidence of loss, breakup, or loosening of any component in the gun bore or in flight when tested at excess and service pressures.

3.7 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

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* 3.7.1 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings and shall be in accordance with the provisions of MIL-A-2550.

3.7.2 Burr.-No component shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.

a. Successful passage of all inspections required by the drawings and specifications shall be considered as a condition precedent to performing the proving ground functional tests (See 3.6).

b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the proving ground functional tests.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning or regular production, a sample shall be submitted in accordance with contract requirements and shall consist of forty-five projectile metal parts as follows:

35 metal parts assemblies complete, including paint
5 metal parts assemblies complete, except paint

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4.2.1 (Cont'd)

5 sets of unassembled component parts complete, including protective coating except those components requiring protective finish in accordance with MIL-STD-171, finish No. 5.1.1 shall be furnished without protective coating.
5 body and band assemblies completely machined, less protective coating.

Thirty each of the above metal parts assemblies complete shall be forwarded to a designated proving ground for functional tests. The sample shall be manufactured in the same manner using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After the inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Forging lots.-Forging lots shall contain not more than one heat of steel.

4.3.1.3 Plate and bar stock lots.-Plate and bar stock lots shall consist of components machined from bar or plate stock and shall contain not more than one heat of steel.

4.3.1.4 Cold shaping lots.-Cold shaping lots shall contain not more than one heat of steel.

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4.3.1.5 Heat treatment lots.-Heat treatment lots shall consist of components manufactured from forgings in accordance with 4.3.1.2, or bar or plate stock in accordance with 4.3.1.3 that have been heat treated in the same furnace at the same temperature in one unchanged process. After final heat treatment, components shall remain grouped for identity purposes until such time as the results of the tension test hereinafter prescribed indicates compliance with mechanical properties of the applicable component drawing.

4.3.1.6 Stress relieved lots.-Stress relieved lots shall consist of components manufactured from cold shapings in accordance with 4.3.1.4 that have been stress relieved in the same furnace at the same temperature in one unchanged process. After stress relief the cold shapings shall remain grouped for identity purposes until such time as the tension test hereinafter prescribed indicates compliance with mechanical properties requirements of applicable component drawing.

4.3.1.7 Projectile lot.-Heat treated, stress relieved or plate and bar stock lots of components that have complied with the mechanical properties requirements of the applicable component drawing, may be regrouped for assembly into projectile lots. A projectile lot shall consist of projectile assemblies produced by the one manufacturer, under one contract, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for roughness determination.

* 4.3.2.1 Windshield, Drawing D10534391, covering a detail of Drawing F10534387.

Defect	Method of Inspection
CRITICAL	
None defined	

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Straightness of ogive	Gage
102. Pitch diameter of thread, max.	Gage
103. Minor diameter of thread, max.	Gage
104. Diameter, outside rear	Gage
105. Diameter forward of thread	Gage
106. Length of thread, min.	Gage
107. Location of forward basic diameter	Gage
108. Location of rear basic diameter	Gage
109. Wall thickness variation	Gage
110. Wall thickness (front end)	Gage
111. Wall thickness (side wall)	Gage
112. Thread(s) damaged or not full	Visual
113. Metal defective	Visual
MINOR	
201. Length to thread run out, max.	Gage
202. Length, total	Gage
203. Radii or chamfer missing or incorrect	Visual (See 6.2)
204. Surface finish improper	Visual (See 6.2)
205. Burr	Visual
206. Foreign matter	Visual

* 4.3.2.2 Fuze Plug, Drawing C10551895, covering a detail of Drawing F10534387.

CRITICAL	
None defined	
MAJOR	
101. Diameter, rear end, not functionally concentric with pitch diameter of thread	Gage
102. Diameter, rear end	Gage
103. Pitch diameter thread, min.	Gage
104. Major diameter of thread, min.	Gage
105. Length from rear end to thread	Gage
106. Thread(s) damaged or not full	Visual
107. Metal defective	Visual

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Length, total	Gage
202.	Radii or chamfer missing or incorrect	Visual (See 6.2)
203.	Surface finish improper	Visual (See 6.2)
204.	Presence and location of wrench holes	Visual (See 6.2)
205.	Protective coating inadequate or damaged	Visual
206.	Burr	Visual

* 4.3.2.3 Spike, Drawing D10534390, covering a detail of Drawing F10534387.

CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of stem with pitch diameter of body thread, including perpendicularity of rear face of spike	Gage
102.	Stem not functionally concentric with cavity	Gage
103.	Rear face not functionally perpendicular with pitch diameter of body thread	Gage
104.	Cavity not functionally concentric with pitch diameter of fuze thread	Gage
105.	Straightness of rear face	Gage
106.	Pitch diameter of body thread, min.	Gage
107.	Major diameter of body thread, min.	Gage
108.	Pitch diameter of windshield thread, min.	Gage
109.	Major diameter of windshield thread, min.	Gage
110.	Pitch diameter of fuze plug thread, max.	Gage
111.	Minor diameter of fuze plug thread, max.	Gage
112.	Length of effective fuze plug thread, min.	Gage
113.	Length from rear face to face of shoulder forward of front windshield thread undercut, max.	Gage
114.	Length from rear face to shoulder forward of body thread, max.	Gage
115.	Length of body thread, min.	Gage
116.	Length of windshield thread, min.	Gage

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
117.	Length of body thread pilot, max.	Gage
118.	Length of windshield thread pilot, max.	Gage
119.	Thread(s) damaged or not full	Visual
120.	Metal defective	Visual

MINOR

201.	Diameter of cavity	Gage
202.	Diameter of stem	Gage
203.	Diameter of body thread undercut, max.	Gage
204.	Diameter of windshield thread undercut, max.	Gage
205.	Depth of cavity	Gage
206.	Length, total	Gage
207.	Presence and location of wrench holes	Visual (See 6.2)
208.	Radii or chamfer missing or incorrect	Visual (See 6.2)
209.	Surface finish improper	Visual (See 6.2)
210.	Burr	Visual
211.	Foreign matter	Visual
212.	Protective coating inadequate or damaged	Visual

* 4.3.2.4 Plate, Drawing C10534392, covering a detail of Drawing F10534387.

CRITICAL

None defined

MAJOR

101.	Rear face not functionally perpendicular with pitch diameter of thread	Gage
102.	Parallelism of front face with rear face	Gage
103.	Straightness of front face	Gage
104.	Pitch diameter of thread, min.	Gage
105.	Major diameter of thread, min.	Gage
106.	Width of plate	Gage
107.	Thread(s) damaged or not full	Visual
108.	Metal defective	Visual

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4.3.2.4 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Diameter of breather hole, max. (if applicable)	Gage
202.	Depth of center drill, max. (if applicable)	Gage
203.	Presence and location of wrench holes	Visual (See 6.2)
204.	Radii or chamfer missing or incorrect	Visual (See 6.2)
205.	Surface finish improper	Visual (See 6.2)
206.	Burr	Visual
207.	Foreign matter	Visual

* 4.3.2.5 Body, Drawing F10534388, covering a detail of Drawing F10534387.

CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of rotating band seat with bourrelets	Gage
102.	Diameter of rotating band seat	Gage
103.	Depth of scores in band seat ribs	Visual (See 6.2)
104.	Metal defective	Visual
MINOR		
201.	Location of rotating band seat	Gage
202.	Width of rotating band seat	Gage
203.	Burr	Visual
204.	Foreign matter in rotating band seat	Visual
205.	Evidence of shot or sand blasting of band seat after knurling	Visual

* 4.3.2.6 Adapter, Tracer, Drawing C10535917, covering a detail of Drawing F10534387.

CRITICAL		
	None defined	

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Tracer cavity not functionally concentric with major diameter of body thread	Gage
102. Body seat not functionally perpendicular with pitch diameter of body thread	Gage
103. Pitch diameter of body thread, min.	Gage
104. Major diameter of body thread, min.	Gage
105. Diameter of body thread undercut, max.	Gage
106. Diameter of tracer thread undercut, min.	Gage
107. Length from rear of tracer adapter to tracer cavity shoulder	Gage
108. Length from front face to body seat, max.	Gage
109. Length of effective body thread, min.	Gage
110. Width of tracer thread undercut, max.	Gage
111. Thread(s) damaged or not full	Visual
112. Metal defective	Visual
MINOR	
201. Diameter of tracer cavity	Gage
202. Diameter, outside	Gage
203. Length from tracer thread undercut to bottom of tracer cavity	Gage
204. Length, total	Gage
205. Presence and location of wrench holes	Visual (See 6.2)
206. Radii or chamfers missing or incorrect	Visual (See 6.2)
207. Surface finish improper	Visual (See 6.2)
208. Protective coating inadequate or damaged	Visual
209. Burr	Visual
210. Foreign matter	Visual

* 4.3.2.7 Ring, Case Locking, Drawing C10534384, covering a detail of Drawing F10534387.

CRITICAL

None defined

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4.3.2.7 (Cont'd)

Defect		Method of Inspection
MAJOR		
101.	Pitch diameter of thread, max.	Gage
102.	Minor diameter of thread, max.	Gage
103.	Thread(s) damaged or not full	Visual
104.	Metal defective	Visual
MINOR		
201.	Pitch diameter of thread, min.	Gage
202.	Minor diameter of thread, min.	Gage
203.	Diameter, outside	Gage
204.	Depth of flat	Gage
205.	Location of flat	Gage
206.	Width of ring	Gage
207.	Width across flats	Gage
208.	Protective coating inadequate or damaged	Visual
209.	Radii or chamfers missing or incorrect	Visual (See 6.2)
210.	Surface finish improper	Visual (See 6.2)
211.	Burr	Visual
212.	Foreign matter	Visual

* 4.3.2.8 Plug and Disc Assembly, Drawing B8595462, covering a detail of Drawing F10534387.

CRITICAL		
None defined		
MAJOR		
101.	Inside diameter not functionally concentric with major diameter of thread	Gage
102.	Pitch diameter of thread, min.	Gage
103.	Major diameter of thread, min.	Gage
104.	Diameter of thread chamfer	Gage
105.	Width of threaded diameter	Gage
106.	Thickness of disc	Gage
107.	Thread(s) damaged or not full	Visual
108.	Improper assembly	Visual
109.	Metal defective	Visual

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Major diameter of thread, max.	Gage
203. Diameter of through hole	Gage
204. Presence and location of wrench holes	Visual (See 6.2)
205. Surface finish improper	Visual (See 6.2)
206. Burr	Visual
207. Foreign matter	Visual
208. Protective coating inadequate or damaged	Visual

* 4.3.2.9 Body and Band Assembly, Drawing D10534389, covering a detail of Drawing F10534387.

CRITICAL

None defined

MAJOR

101. Concentricity of forward cavity with bourrelets	Gage
102. Concentricity of rotating band with bourrelets	Gage
103. Major diameter of case locking ring thread not functionally concentric with minor diameter of tracer adapter thread	Gage
104. Perpendicularity of tracer adapter seat with pitch diameter of tracer adapter thread	Gage
105. Perpendicularity of front face of body with bourrelets	Gage
106. Perpendicularity of rear face of body with bourrelets	Gage
107. Pitch diameter of plate and spike thread, max.	Gage
108. Minor diameter of plate and spike thread, max.	Gage
109. Pitch diameter of tracer adapter thread, max.	Gage
110. Minor diameter of tracer adapter thread, max.	Gage
111. Diameter of bourrelets, min.	Gage
112. Diameter, outside forward end of body	Gage
113. Diameter of rotating band	Gage
114. Length from front face to plate seat	Gage
115. Length of effective tracer adapter thread, min.	Gage
116. Length of effective case locking ring thread, min.	Gage

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4.3.2.9 (Cont'd)

	Defect	Method of Inspection
117.	Length from tracer adapter seat to bottom of rear cavity	Gage
118.	Length from rear of body to forward inside basic diameter of rear cavity	Gage
119.	Length from rear of body to rear inside basic diameter of rear cavity	Gage
120.	Length to basic diameter on rotating band	Gage
121.	Width of rotating band	Gage
122.	Thread(s) damaged or not full	Visual
MINOR		
201.	Diameter of counter bore, front end	Gage
202.	Diameter of cavity to the rear of plate and spike thread undercut	Gage
203.	Diameter of plate and spike thread undercut	Gage
204.	Diameter of tracer adapter thread through hole	Gage
205.	Diameter of body	Gage
206.	Length of tracer adapter thread, max.	Gage
207.	Length, total	Gage
208.	Length from front face to rear of forward bourrelet	Gage
209.	Length from front face to front of rear bourrelet	Gage
210.	Length from rear face to rear basic diameter inside forward cavity	Gage
211.	Length from front face to forward basic diameter inside forward cavity	Gage
212.	Length from rear face to bottom of rear cavity	Gage
213.	Location of rotating band	Gage
214.	Radii or chamfer missing or incorrect	Visual (See 6.2)
215.	Surface finish improper	Visual (See 6.2)
216.	Burr	Visual
217.	Foreign matter	Visual

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* 4.3.2.10 Metal Parts Assembly, F10534387 (prior to painting).

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of forward end of windshield with bourrelets	Gage
102. Concentricity of tracer adapter with bourrelets	Gage
103. Concentricity of body with bourrelets	Gage
104. Windshield diameter flush or below bourrelet diameter	Gage
105. Length from rear face of body to front face of spike, max.	Gage
106. Insecure assembly (tracer adapter to body)	Torque Wrench (See 4.4.3)
107. Insecure assembly (plate to body)	Torque Wrench (See 4.4.3)
108. Insecure assembly (spike to body)	Torque Wrench (See 4.4.3)
109. Insecure assembly (windshield to spike)	Torque Wrench (See 4.4.3)
110. Insecure assembly (fuze plug to spike)	Torque Wrench (See 4.4.3)
111. Gap between windshield and body, max.	Gage
MINOR	
201. Adhesive compound on exterior surfaces	Visual
202. Marking missing, incorrect or illegible	Visual
203. Burr	Visual
204. Foreign matter	Visual
205. Protective coating inadequate or damaged	Visual

* 4.3.2.11 Metal Parts Assembly, Drawing F10534387 (after painting).

CRITICAL

None defined

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4.3.2.11 (Cont'd)

Defect		Method of Inspection
MAJOR		
101.	Weight	Approved Scale
102.	Pitch diameter of case locking ring thread, min.	Gage
103.	Major diameter of case locking ring thread, min.	Gage
104.	Pitch diameter of tracer thread, max.	Gage
105.	Minor diameter of tracer thread, max.	Gage
106.	Diameter of bourrelets, max.	Gage
107.	Thread(s) damaged or not full	Visual
108.	Rotating band damaged	Visual
MINOR		
201.	Pitch diameter of case locking ring thread, max.	Gage
202.	Major diameter of case locking ring thread, max.	Gage
203.	Pitch diameter of tracer thread, min.	Gage
204.	Minor diameter of tracer thread, min.	Gage
205.	Paint not dry	Tactile
206.	Paint on thread(s)	Visual
207.	Protective coating inadequate or damaged	Visual
208.	Marking missing, incorrect or illegible	Visual
209.	Burr	Visual
210.	Foreign matter	Visual

* 4.3.2.12 Pallet, Drawing F8889328.

CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201.	Pallet damaged (unserviceable)	Visual
202.	Sleeve(s) missing	Visual
203.	Improperly packed	Visual
204.	Strapping missing, loose or broken	Visual/Manual
205.	Marking missing, incorrect or illegible	Visual

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4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Hardness.

4.3.3.1.1.1 Heat treatment lots.-In order to determine uniformity of heat treatment, each component in the heat treatment lot (as defined by 4.3.1.5) shall be subjected to a hardness test.

4.3.3.1.1.2 Stress relieved lots.-From each stress relieved lot (as defined by 4.3.1.6), the first twenty-five components processed through the stress relief furnace and five from each one hour production thereafter, for the balance of the stress relieved lot shall be hardness tested to assess uniformity of process. Records of hardness results shall be maintained.

4.3.3.1.2 Tension test.

4.3.3.1.2.1 Heat treatment lots.-Two components from each heat treatment lot (one of the hardest and one of the softest as determined by 4.3.3.1.1.1) shall be subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of one or more tensile test specimens to comply with the yield strength and elongation requirements shall be cause for rejection of the lot with the exception that elongation results obtained from tangentially transverse specimens in the base web area of the body shall be recorded for informational purposes only.

4.3.3.1.2.2 Stress relieved lots.-Two components from each stress relieved lot shall be subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of one or more tensile test specimens to comply with the yield strength and elongation requirements shall be cause for rejection of the lot with the exception that elongation results obtained from tangentially transverse specimens in the base web area of the body shall be recorded for informational purposes only.

4.3.3.1.2.3 Plate and bar stock lots.-With the approval of the procuring activity when components are manufactured from plate or bar stock in such a manner that the mechanical properties are not altered, three plates or bars as applicable from each plate or bar stock lot may be tested in lieu of the component. Failure of any tensile test specimen to comply with the yield strength and elongation requirements of the applicable drawing shall be cause for rejection of the plate or bar stock lot.

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4.3.3.2 Aluminum components.-The test procedure, sample size and acceptance criteria shall be as specified in the applicable material specification. Failure to comply with the requirements of the applicable drawing and specification shall be cause for rejection of the material represented by the sample.

4.3.3.3 Rotating band.

4.3.3.3.1 Band seating.-Two body and band assemblies from each lot, prior to painting, shall be subjected to this test to determine compliance with the applicable drawing. Failure of either assembly to comply shall be cause for rejection of the lot.

4.3.3.3.2 Density.-The samples provided by 4.3.3.3.1, after examination for seating, shall be subjected to this test to determine compliance with the applicable drawing. Failure to comply shall be cause for rejection of the lot.

4.3.3.4 Torque test.

4.3.3.4.1 Tracer adapter to body.-The body and tracer adapter shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of the sample to comply with the criteria of the sampling plan as evidenced by a perceptible movement shall be cause for rejection of the lot.

4.3.3.4.2 Plate to body.-The body and plate shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of the sample to comply with the criteria of the sampling plan as evidenced by a perceptible movement shall be cause for rejection of the lot.

4.3.3.4.3 Spike to body.-The body and spike shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of the sample to comply with the criteria of the sampling plan as evidenced by a perceptible movement shall be cause for rejection of the lot.

* 4.3.3.4.4 Fuze plug to spike.-The fuze plug shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of the sample to comply with the criteria of the sampling plan as evidenced by a perceptible movement shall be cause for rejection of the lot.

* 4.3.3.4.5 Windshield to spike.-The windshield shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of the sample to comply with the criteria of the sampling plan as evidenced by a perceptible movement shall be cause for rejection of the lot.

4.3.3.5 Cleaning and application of protective coating.-The sample size shall be as specified in the applicable specification with the exception that the sample size for the corrosive resistance test for those components requiring protective finish in accordance with MIL-C-60539, Type II, Class 1 shall be selected in accordance with MIL-STD-105, Inspection Level S-2. The acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements of the applicable drawings and specifications shall be cause for taking action prescribed in the applicable specification.

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* 4.3.3.6 Tool control dimensions.-Dimensions designated as tool control type dimensions shall be gaged at the beginning of each eight hours production and whenever tooling is replaced. When destruction of component is necessary to inspect these dimensions, measurement of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool.

4.3.4 Proving ground functional tests.-The following proving ground tests will be performed by the Government.

4.3.4.1 First article sample.

4.3.4.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated.

TABLE I

Test Phase	Quantity
Excess Pressure	15
Service Pressure	15

4.3.4.2 Production lot sample.-The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

Test Phase	Quantity
Excess Pressure	10
Service Pressure	10 *

* After ten (10) successive lots have been tested without a single failure, the sample size may be reduced to five (5) metal parts assemblies from each production lot.

4.3.4.3 Excess pressure.-Samples provided in accordance with Tables I and II shall be subjected to this test. Failure of any projectile assembly to comply with 3.6 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.4.4 Service pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.6 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

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4.3.5 Inspection equipment.-Index of Inspection Equipment List IEL-10534387 identifies the inspection equipment required to perform the examinations and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Hardness.-After final heat treatment or stress relief and before final machining, the hardness test shall be performed on the body at the location of the band seat and on the rear outside surface of the windshield. The tests shall be made using the procedure for performing this test as prescribed by Federal Test Method Standard No. 151.

4.4.1.2 Tension test.

* 4.4.1.2.1 Yield strength and elongation (heat treatment and stress relieved lots).-From the components selected in accordance with 4.3.3.1.2.1 and 4.3.3.1.2.2, tension test specimens shall be selected from those components requiring heat treatment as follows: From each body, one test specimen shall be taken longitudinally from midpoint in the body and one test specimen shall be taken from each body in a tangentially transverse direction from the central portion of the base web area. The centerline of this specimen shall be located 2 inches plus 1/4 inch from the axis of the body. The body specimens shall conform to the largest obtainable standard "R" type specimen. From each windshield, two full length test specimens shall be taken longitudinally 180 degrees apart. They shall conform to the largest obtainable standard "T" type specimen. From each plate, one tangentially transverse test specimen shall be taken, located approximately 1½ inches from the axis of the component. They shall conform to the largest obtainable standard "R" type specimen. All specimen types shall conform to Federal Test Method Standard No. 151. Yield strength shall be determined by the offset method utilizing the procedures of Federal Test Method Standard No. 151.

4.4.1.2.2 Yield strength and elongation (plate and bar stock lots).-From the end of each bar selected in accordance with 4.3.3.1.2.3, one longitudinal tensile test specimen shall be removed. It shall be located at mid-radius of the bar and shall conform to the largest obtainable standard "R" type specimen. One tensile test specimen shall be removed from each plate selected in accordance with 4.3.3.1.2.3. It shall conform to the largest obtainable standard "R" type specimen. All specimens shall conform to Federal Test Method Standard No. 151. Yield strength shall be determined by the offset method utilizing the procedures of Federal Test Method Standard No. 151.

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4.4.1.3 Aluminum components.-Procedures for performing the mechanical properties tests shall be in accordance with the criteria prescribed by the applicable material specification.

4.4.2 Rotating band.

4.4.2.1 Band seating.-From the sample provided by 4.3.3.3, the body and band assembly shall be carefully sectionalized with a wet abrasive cut-off wheel, so as to produce four segments of 90 degrees each. Visual examination of the exposed surfaces of the rotating band shall then be conducted to determine compliance with the applicable drawing.

4.4.2.2 Density.-After examination for band seating as prescribed in 4.4.2.1, three specimens from each body and band assembly spaced approximately 120 degrees apart, shall be removed from the rotating bands. The bands shall be the full width and thickness of the band and approximately one inch long. The specimens shall be sawed by hand to the specified size. With the exception of removing burrs, no further machining shall be performed on the specimens. After weighing the specimens, the wax shall be removed by either one of the two methods described in "a" and "b".

a. Place specimens in a dry hydrogen atmosphere furnace and hold at 1400 degrees F. for one hour. Remove specimens and weigh. Return specimens to the furnace and hold at 1400 degrees F. for 30 minutes. This procedure with the half hour heating intervals shall be repeated until the change in weight is less than .05%.

* b. A soxhlet apparatus of suitable size using either toluene or trichloroethylene may be used for extracting the wax. The extraction time shall be a minimum of 18 hours. The specimens shall be dried for 1½ hours in an oven at 210 degrees F. and then weighed. This procedure with 1½ hour extraction intervals shall be repeated until the change in weight is less than .05%.

Following wax removal and weighing, the specimens shall be impregnated with micro-crystalline wax. This shall be accomplished by immersing the specimen in molten micro-crystalline wax held at a temperature of 180 degrees F. to 200 degrees F. The pressure shall then be reduced by a suitable evacuating method to not more than 2 inches of mercury and held for 30 minutes. The pressure shall then be slowly increased (approximately 30 seconds) to atmospheric pressure after which the specimen shall remain immersed in the molten micro-crystalline wax at 180 degrees F. to 200 degrees F. for approximately one hour.

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The density of the specimens shall be calculated as follows:

$$D = \frac{A}{B - C}$$

Where

D = density in grams per cubic centimeter
A = weight in air of wax free specimen, in grams
B = weight in air of micro-crystalline impregnated specimen, in grams
C = weight of micro-crystalline wax impregnated specimen immersed in water, in grams

4.4.2.3 Density and wax content of band blank.-The procedure for determining density and wax content shall be as specified in 4.4.2.2 except that from the band blanks selected in accordance with 4.3.3 and 4.3.4 of MIL-R-11073, nine specimens shall be sectioned from each band. They shall consist of three sets spaced approximately 120 degrees apart. The approximate size of each specimen shall be one inch long x 1/3 the width of the band x the full thickness of the band (see Figure 1). The specimens shall be sawed by hand to the specified size. With the exception of removing burrs, no further machining shall be performed.

4.4.3 Torque test.

4.4.3.1 Tracer adapter to body.-The assemblies selected in accordance with 4.3.3.4.1 shall be mounted in a suitable holding device in such a manner that the body component is prevented from turning. The applicable torque shall be applied to the tracer adapter in the direction of assembly. Testing shall be conducted while thread sealant is wet.

4.4.3.2 Plate to body.-The assemblies selected in accordance with 4.3.3.4.2 shall be mounted in a suitable holding device in such a manner that the body component is prevented from turning. The applicable torque shall be applied to the plate in the direction of assembly. Testing shall be conducted while thread sealant is wet.

4.4.3.3 Spike to body.-The assemblies selected in accordance with 4.3.3.4.3 shall be mounted in a suitable holding device in such a manner that the body component is prevented from turning. The applicable torque shall be applied to the spike in the direction of assembly. Testing shall be conducted while thread sealant is wet.

* 4.4.3.4 Fuze plug to spike.-The assemblies selected in accordance with 4.3.3.4.4 shall be mounted in a suitable holding device in such a manner that the spike component is prevented from turning. The applicable torque shall be applied to the fuze plug in the direction of assembly. Testing shall be conducted while thread sealant is wet.

* 4.4.3.5 Windshield to spike.-The assemblies selected in accordance with 4.3.3.4.5 shall be mounted in a suitable holding device in such a manner that the body component is prevented from turning. The applicable torque shall be applied to the spike in the direction of assembly. Testing shall be conducted while thread sealant is wet.

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4.4.4 Protective coating.-The procedure for testing the cleaning of components and the application of protective coating shall be in accordance with the criteria for testing as specified in the applicable specification.

* 4.4.5 Proving ground functional tests.

* 4.4.5.1 Excess pressure.-The metal parts assemblies supplied in accordance with 4.3.4.3 will be assembled into complete rounds and fired in a gun for which the projectile is standard, the tube of which will have 50 to 25 percent life remaining based on current gun tube condemnation criteria. The propellant charge shall be adjusted to produce 112 percent of the standard service pressure of the round in a new gun. The projectile will be fired through a target placed 2000 meters from the muzzle of the gun. Photographic equipment will be set up near the muzzle of the gun in such a manner as to record the flight of the projectile prior to its impact with the target. Photographs will be examined for compliance with requirements of 3.6.

* 4.4.5.2 Service pressure.-The metal parts assemblies supplied in accordance with 4.3.4.4 will be inert loaded and assembled into complete rounds, utilizing the service charge. The rounds will be fired in a gun for which the projectile is standard, the tube of which shall have a remaining service life of 50 percent based on current gun tube condemnation criteria. The test rounds will be fired alternately with projectiles from a reference lot for comparison. All rounds will have velocities measured and recorded (for informational purposes, see 6.1) and will be fired at a vertical target placed 2000 meters from the gun muzzle. Accuracy in terms of maximum spread in inches both horizontal and vertical and target dispersion in terms of standard deviation in mils shall be recorded for all rounds fired (for informational purposes, see 6.1). Photographic equipment will be set up near the gun muzzle in such a manner as to record the flight of the projectile prior to its impact with the target. Photographs will be examined for compliance with requirements of 3.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing.

5.2 Packing.

5.2.1 Level A.-Not applicable.

5.2.2 Level B.-Not applicable.

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5.2.3 Level C.-Minimum military pack.-The metal parts for Projectile, TP-T, 152MM, XM411E1, shall be packed as follows for shipment to the designated loading facility:

5.2.3.1 Metal parts assembly.-Part No. 10534387 shall be packed in accordance with Drawing F8889328.

5.2.3.2 Plug and disc assembly.-Part No. 8595462 shall be packed in accordance with MIL-STD-1169. Pack 2500 plug and disc assemblies arranged in a 10 x 10 pattern, 25 to a cell, using interlocking separators.

5.2.3.3 Ring, case locking.-Part No. 10534384 shall be packed in accordance with MIL-STD-1169. Pack 400 rings, case locking arranged in a 4 x 5 pattern, two layers, 20 to a cell, using interlocking separators.

5.3 Shipping.-When component parts from more than one lot are shipped as a carload or truckload, each lot must be kept separate and the devision between lots clearly indicated.

5.4 Marking.-The pallet shall be marked in accordance with Drawing F8889328. Cartons shall be marked in accordance with MIL-STD-1169.

5.5 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Velocity-accuracy.-Proving Ground tests for velocity-accuracy are not acceptance tests, but the results will be carefully analyzed by the Munitions Command to determine quality of ammunition produced. Evidence of excessive variation in any phase of the Proving Ground tests will result in immediate investigation by the Munitions Command.

6.2 Visual examination qualifications.-When compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.3 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

* The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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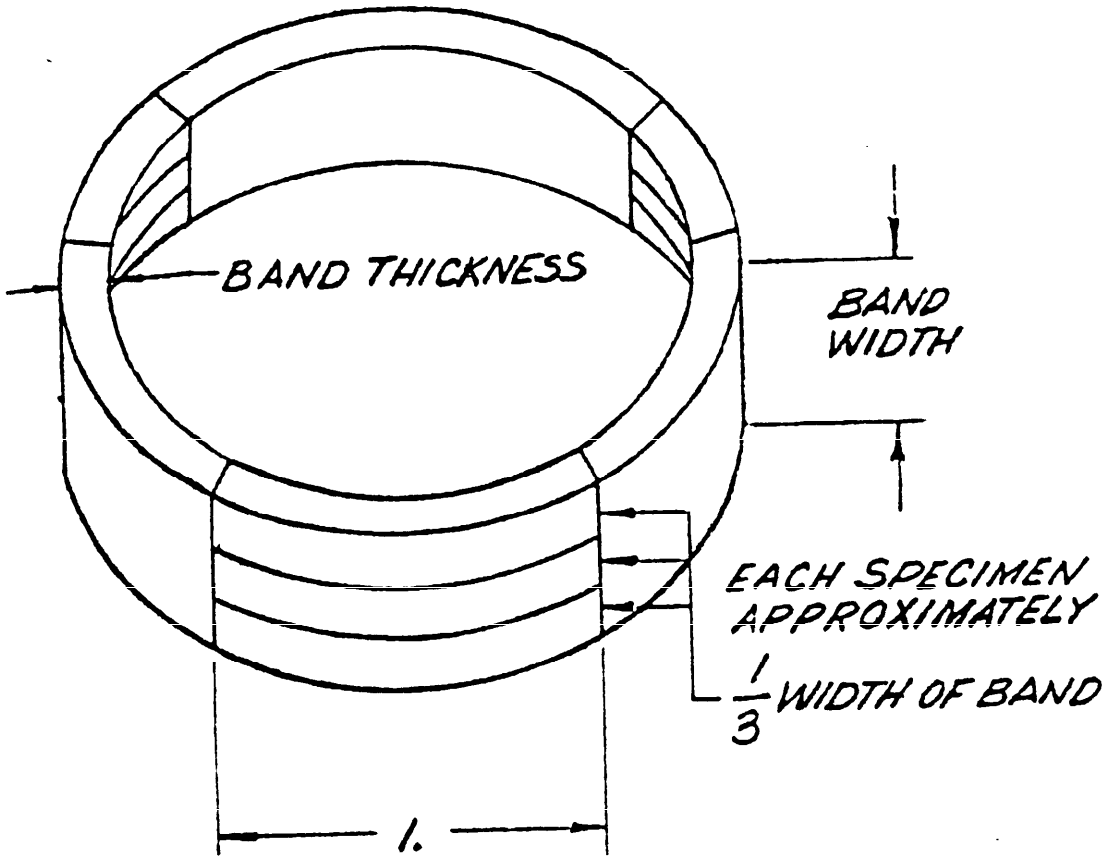
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Army - MU

Project No. 1320 -A100

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THREE SETS OF SPECIMENS AS INDICATED, SHALL BE SECTIONED FROM EACH BAND TO BE TESTED, SPACED APPROXIMATELY 120° APART.

FIG. 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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